

Xiao Fu

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EDUCATION

Zhejiang University

Hangzhou, China

Bachelor in Information Engineering | College of Information Science & Electronic Engineering Sept. 2018 – Jul. 2022

- **GPA:** 3.95/4.00, 90.53/100, **Ranking:** 4/145 (Top 3%)
- **Two-time National Scholarship winner, Dean's Honor Graduate**
- **Honors Program:** Zhejiang University Morningside Cultural China Scholars Program (36 students selected from 12,000 freshmen with distinguished innovation capability and leadership)

PUBLICATIONS

* equal contribution, † corresponding author

- [2] [Xiao Fu*](#), Shangzhan Zhang*, Tianrun Chen, Yichong Lu, Lanyun Zhu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao†. “**Panoptic NeRF: 3D-to-2D Label Transfer for Panoptic Urban Scene Segmentation**”. In *Proceedings of the International Conference on 3D Vision (3DV)*, 2022 [[Project Page](#)[Paper](#)[Code](#)]
- [1] Guangyi Zhang*, [Xiao Fu*](#), Qiyu Hu†, Yunlong Cai, Guanding Yu. “**Hybrid Precoding Design Based on Dual-Layer Deep-Unfolding Neural Network**”. *IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Helsinki, Finland, 2021, pp. 678-683 [[Paper](#)[Code](#)]

RESEARCH EXPERIENCE

Panoptic NeRF: 3D-to-2D Panoptic Label Transfer

Max Planck Institute for Intelligent Systems

Supervisor: [Prof. Andreas Geiger](#) and [Prof. Yiyi Liao](#)

Sept. 2021 – Mar. 2022

- Propose Panoptic NeRF, which aims for obtaining per-pixel 2D semantic and instance labels from easy-to-obtain coarse 3D bounding primitives. Our method utilizes NeRF as a differentiable tool to unify coarse 3D annotations and 2D semantic cues transferred from existing datasets. This combination allows for improved geometry guided by semantic information which enables rendering of accurate semantic maps across multiple view. By inferring in 3D space and rendering to 2D labels, our 2D semantic and instance labels are multi-view consistent by design

Algorithm-Based Dual-Layer Deep-Unfolding Neural Network

Zhejiang University

Supervisor: [Prof. Guanding Yu](#)

Jun. 2020 – Feb. 2021

- Proposed a novel framework, i.e. a Dual-Layer Deep-Unfolding Neural Network (DLDUNN) which unfolds the iterative Penalty Dual Decomposition (PDD) algorithm into a layer-wise structure whereby some trainable parameters are introduced into the places of the high-complexity operations, e.g. large-dimensional matrix inversions and iterative sub algorithms, so as to address the high computational complexity problem of the dual-layer PDD algorithm and presents the major original performance in the meantime
- **Laureate of Excellent Award (Team leader) - Student Research Training Program (Province-level)**

HONORS AND SCHOLARSHIPS

- National Scholarship (**Highest Honor for Chinese Undergraduates, Twice**) 2020,2021
- ISEE Excellent Student Award (**Top 5 among ISEE 314 undergraduates, Dean's Honor**) 2021
- ISEE Yuelun Alumni Scholarship (**Dean's Award for Academic Contest, Bonus 20,000 CNY**) 2020,2021
- First-Class Scholarship for Academic Excellence of Zhejiang University (**Top 3%**) 2020,2021
- First Prize of the National Talents Training Base (**Top 3%**) 2020
- Government Scholarship for Academic Excellence of Zhejiang Province (**Top 3%**) 2019
- Outstanding Graduation Thesis Award of Zhejiang University 2022
- Finalist Winner – Mathematical Contest in Modeling (MCM/ICM) (**Top 2% out of 20,954 teams**) 2020
- Meritorious Winner – Mathematical Contest in Modeling (MCM/ICM) (**Top 7% out of 26,112 teams**) 2021
- Gold Prize – 7th “Internet Plus” College Student Innovation and Entrepreneurship Contest (**1st author**) 2021
- Gold Prize – 12th “Challenge Cup” National College Student Business Plan Competition, National 2020
- Grand Prize – “Yongdian Cup” Innovation and Entrepreneurship Contest (**1st author**) 2019
- Third Prize – 17th “Challenge Cup” Extracurricular Academic Science and Technology Contest (**1st author**) 2020
- Outstanding Student Award of Zhejiang University 2019,2020,2021
- Academic Excellence Scholarship of Zhejiang University 2019,2020,2021
- Zhejiang University Five-Star Volunteer 2021